

# **Legacy Procedure Remediation - Hydraulic Test Stand HTS-100**

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N. B. Smith - writing sample

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# Legacy procedure remediation

## Title page and release statement

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### Release statement

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This publication contains none of the following:

- Classified information of any level.
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- Export-controlled technical data under the International Traffic in Arms Regulations (ITAR), 22 CFR 120-130.
- Export-controlled technical data under the Export Administration Regulations (EAR), 15 CFR 730-774.

All content is original work. The equipment in this publication, the Hydraulic Test Stand HTS-100, does not exist. The legacy procedure in data module 040A is a synthetic example that the author wrote to show typical defects. It is not an extract from a real publication, and it is not subject to the copyright of any other party.

Prepared by Nathan B. Smith.

### Purpose of this publication

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This publication shows the remediation of a legacy maintenance procedure. It has three parts:

1. The legacy procedure as found, with the defects identified.
2. The procedure after the rewrite, as an S1000D procedural data module.
3. The rationale for each class of change.

The rewrite uses Simplified Technical English in accordance with ASD-STE100.

| NOTE                                                                                                                                        |
|---------------------------------------------------------------------------------------------------------------------------------------------|
| This publication is a technical writing sample. Do not use it as a maintenance instruction. Use the approved publication for the equipment. |

# Hydraulic test stand HTS-100

Legacy servicing procedure as found

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## Source text

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The text that follows is the legacy procedure without a change. The numbers in brackets identify the defects. Data module 040B gives the analysis of each defect.

### 3.4.2 RESERVOIR SERVICING

3.4.2.1 The reservoir should be serviced at 250 hour intervals or whenever the fluid level is observed to be low. It is important that only approved fluid be used. [1] [2] [3]

3.4.2.2 Prior to commencing the servicing operation the technician will ensure that the system has been depressurized and that the unit has been allowed to cool, after which the access panel may be removed by removing the eight (8) fasteners securing same and the filter element should be inspected for contamination and replaced if necessary. [4] [5] [6]

3.4.2.3 Fluid is added through the fill port until it reaches the FULL mark on the sight glass. Do not overfill it. [7] [8]

3.4.2.4 NOTE: Hydraulic fluid under pressure can penetrate the skin and cause serious injury. Personnel should exercise caution. [9] [10]

3.4.2.5 Reinstall the panel. The fasteners are torqued to 45 in-lbs. Return the unit to service and check for leaks. [11] [12]

## Defect index

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### Defects in the legacy text

| No. | Defect class            | What is wrong                                                                                      |
|-----|-------------------------|----------------------------------------------------------------------------------------------------|
| 1   | Non-instruction         | "should be serviced" does not tell the technician to do anything. It is a statement of preference. |
| 2   | Undefined condition     | "observed to be low" gives no measurable limit.                                                    |
| 3   | Unresolved reference    | "approved fluid" names no specification and no part number.                                        |
| 4   | Compound step           | One sentence contains five separate actions and two conditions.                                    |
| 5   | Absent hazard control   | Depressurization is a safety action, but the text gives it as a subordinate clause.                |
| 6   | Undefined criterion     | "replaced if necessary" gives no accept or reject limit for contamination.                         |
| 7   | Passive voice, no actor | "Fluid is added" does not say who adds it or how.                                                  |
| 8   | Ambiguous pronoun       | "Do not overfill it" - the reservoir, the sight glass, or the fill port.                           |
| 9   | Incorrect hazard type   | A risk of injury to a person is a WARNING, not a NOTE.                                             |
| 10  | Hazard after the action | The hazard statement is after the steps that contain the hazard.                                   |
| 11  | Buried data             | The torque value is in a narrative sentence. The technician cannot find it quickly.                |
| 12  | Absent close-out        | "check for leaks" gives no method, no duration, and no accept criterion.                           |

# Hydraulic test stand HTS-100

## Servicing of the hydraulic reservoir

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### Preliminary requirements

#### Required conditions

- The test stand is shut down.
- The hydraulic system is depressurized to 0 bar (0 psi).
- The reservoir surface temperature is 40 degC (104 degF) or less.
- Electrical power is disconnected and a tag is on the disconnect switch.

#### Personnel

| Trade              | Skill level | Estimated time |
|--------------------|-------------|----------------|
| Hydraulic mechanic | sk02        | 0.5 h          |

#### Support equipment

| Name                                         | Part number | Qty |
|----------------------------------------------|-------------|-----|
| Torque wrench, 2 to 12 N.m (18 to 106 in.lb) | TW-0212     | 1   |
| Fluid transfer pump with 10 micron filter    | FTP-010     | 1   |
| Eye protection and chemical-resistant gloves |             | 1   |

#### Supplies

| Name                           | Part number | Qty |
|--------------------------------|-------------|-----|
| Hydraulic fluid, MIL-PRF-83282 | HF-83282    | 4 L |
| Filter element                 | FE-2930     | 1   |
| Lint-free cloth                |             | 2   |

#### Spares

None.

## Safety conditions

### WARNING

Hydraulic fluid under pressure can go through the skin. An injection injury can cause the loss of a limb, and it gives no pain at the start. Make sure that the system shows 0 bar (0 psi) before you loosen a fitting. Do not use your hand to find a leak.

### WARNING

Hydraulic fluid causes injury to the eyes. Put on eye protection and chemical-resistant gloves before you start this procedure.

### CAUTION

Dirt in the hydraulic fluid causes damage to the pump. Keep the fill port closed except during step 5. Use the transfer pump with the filter. Do not pour fluid from a container into the reservoir.

## Procedure

1. Do the servicing at 250 flight hours, or when the fluid level is below the ADD mark on the sight glass.

2.

### WARNING

A hot surface causes burns to the skin. Make sure that the reservoir surface temperature is 40 degC (104 degF) or less before you touch the reservoir.

Make sure that the pressure gage shows 0 bar (0 psi).

3. Remove the access panel:
  - a. Remove the eight fasteners from the access panel.
  - b. Keep the fasteners in a container.
  - c. Remove the access panel.
4. Examine the filter element:
  - a. Remove the filter element.
  - b. Examine the element for metal particles, for a dark color, and for damage to the pleats.

c.

**NOTE**

Metal particles on the element show wear in the pump. Report this condition before you continue.

If the element has one of these conditions, install a new element, part number FE-2930.

5.

**CAUTION**

Too much fluid in the reservoir causes damage to the seals when the fluid becomes hot. Do not fill the reservoir above the FULL mark on the sight glass.

Add hydraulic fluid MIL-PRF-83282 through the fill port with the transfer pump until the level is at the FULL mark on the sight glass.

6. Clean the fluid from the area around the fill port with a lint-free cloth. Then close the fill port.

7. Install the access panel:

- a. Put the access panel in position.
- b. Install the eight fasteners.
- c. Tighten the fasteners in a crossed sequence to 5.1 N.m (45 in.lb).

8. Do a leak test:

- a. Connect the electrical power. Then remove the tag from the disconnect switch.
- b. Start the test stand. Then increase the pressure to 207 bar (3000 psi).

c.

**WARNING**

Do not use your hand to find a leak. Use a piece of cardboard. Fluid from a small leak can go through the skin.

Hold the pressure for five minutes. Then examine the fittings and the access panel for leaks.

- d. The equipment is serviceable if there is no drop of fluid at the end of the five minutes.

## Requirements after the job

- The fluid level is at the FULL mark with the system at 0 bar (0 psi).
- No leak is present after five minutes at 207 bar (3000 psi).
- The servicing entry is in the equipment record.



# Hydraulic test stand HTS-100

## Rationale for the remediation

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### Principle

Each change in this remediation comes from a rule, not from a preference. A change that comes from a preference cannot be defended in a review, and a different editor makes the opposite change six months later.

### Structural changes

The legacy text has all information in narrative paragraphs. S1000D puts each class of information in its own element. The technician then finds the information without a search of the prose.

#### Movement of information into S1000D structure

| Information in the legacy text                     | New location                | Why                                                                                             |
|----------------------------------------------------|-----------------------------|-------------------------------------------------------------------------------------------------|
| "system has been depressurized", "allowed to cool" | reqCondGroup                | These are conditions before the work, not steps. A technician must know them before he starts.  |
| "only approved fluid"                              | reqSupplies                 | The specification and the part number are now findable before the technician goes to the store. |
| "torqued to 45 in-lbs"                             | reqSupportEquips and a step | The value needs a tool with the correct range. The legacy text names no tool.                   |
| "NOTE: fluid can penetrate the skin"               | reqSafety and step 8        | The hazard applies to the full procedure and again at the leak test.                            |
| "check for leaks"                                  | closeRqmts and step 8       | A close-out condition is a separate class of information from a step.                           |
| Nothing in the legacy text                         | reqPersons                  | The trade, the skill level, and the time are necessary for the maintenance plan.                |

### Hazard changes

The legacy text gives an injury hazard as a NOTE, and it gives the NOTE after the steps that contain the hazard. Both conditions are incorrect.

**Hazard classification rules applied**

| Element | Use it when                                               | Example in this publication                                  |
|---------|-----------------------------------------------------------|--------------------------------------------------------------|
| WARNING | A person can be injured or killed.                        | Fluid injection through the skin. Hot surface.               |
| CAUTION | Equipment can be damaged.                                 | Contamination of the fluid. Too much fluid in the reservoir. |
| NOTE    | Information helps, but no injury and no damage can occur. | Metal particles show wear in the pump.                       |

Each hazard statement is now before the step that contains the hazard. A hazard statement after the action gives no protection, because the technician reads it after he does the work.

The fluid injection warning is in two locations. It is in reqSafety because it applies to the full procedure, and it is again at step 8 because the technician does the leak test many minutes after he read the preliminary requirements.

**Language changes**

The rewrite uses Simplified Technical English. These rules had the most effect on this procedure:

**ASD-STE100 rules applied**

| Rule                                            | Legacy text                             | Rewrite                                          |
|-------------------------------------------------|-----------------------------------------|--------------------------------------------------|
| Use the imperative for an instruction.          | "The reservoir should be serviced"      | "Do the servicing"                               |
| One instruction in one sentence.                | Paragraph 3.4.2.2 has five actions.     | Steps 2 to 4, one action in each.                |
| Use the active voice.                           | "Fluid is added"                        | "Add hydraulic fluid ... with the transfer pump" |
| Do not use an ambiguous pronoun.                | "Do not overfill it"                    | "Do not fill the reservoir above the FULL mark"  |
| Give a measurable limit.                        | "allowed to cool"                       | "40 degC (104 degF) or less"                     |
| Use approved words only.                        | "Prior to commencing", "ensure", "same" | "Before you start", "make sure", "the panel"     |
| Keep a procedural sentence to 20 words or less. | Paragraph 3.4.2.2 has 61 words.         | The longest step has 19 words.                   |

## What the rewrite could not correct

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Three items need data from the responsible engineer. A writer must not invent them:

- The accept and reject limits for filter contamination. The rewrite gives visual indications, but a particle count limit is better.
- The torque sequence for the eight fasteners. The rewrite says "in a crossed sequence". A figure with the numbered sequence is better.
- The action after a leak. The rewrite gives the accept criterion, but it does not give the corrective action.

| NOTE                                                                                                                                                                            |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Record each item of this type as a question to the engineer. Do not close a remediation with an invented value. A gap that is visible is safer than a gap that a writer filled. |